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The emotional cost of AI chatbots in education: Who benefits and who struggles?

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ABSTRACT

Recent advancements in large language models have enabled the development of advanced chatbots, offering new opportunities for personalized learning and academic support that could transform the way students learn. Despite their growing popularity and promising benefits, there is limited understanding of the psychological impact. Accordingly, this study examined the effects of chatbot usage on students' positive and negative affect and considered the moderating role of familiarity. Using a pre-post control group design, undergraduate students were divided into two groups to completed an assignment. Groups received the same task, and only differed based on receiving instruction to use or not to use an AI chatbot. Students who used a chatbot reported significantly lower positive affect, with no significant difference in negative affect. Importantly, familiarity with chatbots moderated changes in positive affect such that students with more familiarity with chatbots reported fewer declines. These findings showcase chatbots' duplicitous effects. While the tools may prove empowering with effective use, they can also decrease the positive aspects of completing assignments for those with less familiarity. These findings underscore the behavioral complexity of AI integration by highlighting how familiarity moderates affective outcomes and how chatbot use may reduce positive emotional engagement without increasing negative affect. Integrating AI tools in education requires not just access and training, but a nuanced understanding of how student behavior and emotional well-being are shaped by their interaction with intelligent systems.

1. Introduction

Artificial Intelligence (AI) has not only changed how students interact with course materials and assignments, but may also alter how students learn moving forward. AI tools provide students with unique support and assistance (Holmes et al., 2023; Luckin, 2017), and of all the advancements, the most disruptive have been large language models. These models are based on sophisticated machine learning algorithms that, when used as chatbots, can generate unique language responses (Pinski & Benlian, 2024). Despite their sophistication, chatbots provide an interactive platform for humans to engage with computers, not in mechanical or coding terms, but in conversational language (Kuhail et al., 2022). As an example, someone could visit any number of publicly available chatbots on the internet and, with a simple prompt (e.g., "revise my cover letter to fit the job description at this link"), receive a very specific document in just a few seconds. Unfortunately, implementing chatbots' use in ways which limit biases and protect ethical human-AI collaborations remains a challenge (García-López et al., 2025).

Among those many challenges exists one which threatens students' wellbeing. More specifically, while the use of chatbots reduces students' workloads overall, their emotional effects remain questioned and underexplored (Labadze et al., 2023). Students recognize the potential of chatbots to provide personalized learning, assist with writing and brainstorming, and enhance research and analysis (Chan & Hu, 2023). With the ability to reduce time and effort on some tasks, more effort can be given elsewhere. This is especially helpful for those with limited resources and other inequities, since chatbots have the potential to offset some of those concerns (Holmes et al., 2023; Luckin, 2017). Despite the promising contributions, the use of chatbots may invite negative outcomes as well. In education specifically, while tutoring and assessment may be optimized (Memarian & Doleck, 2023), the learning process involves more than the transfer of information alone. Indeed, chatbots may enhance academic performance and effectiveness (Zhai et al., 2021), at least to the extent whereby performance stems from information output. Yet, output alone ignores the process of formulation that

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may be critical to students' perceptions of efficacy (Bandura, 1997). That is, being provided the right answer without effort to find the right answer may bypass the process whereby personal growth occurs. This study is situated within the growing field of AI and human behavior by exploring how chatbot usage shapes students' affective responses, contributing to broader debates on cognitive offloading, behavioral adaptation, and technology-mediated learning experiences (Kuhail et al., 2022; Labadze et al., 2023).

On the other hand, spending less time on tasks which could be accomplished by a chatbot could facilitate more time for students to build efficacy in other areas. Therefore, tasks and information aside, students' perceptions during their efforts matter a great deal. Nevertheless, the emotional effect of chatbot usage has received limited attention (Kliene et al., 2024; Memarian & Doleck, 2023). Ortega-Ochoa et al. (2024) and Yan et al. (2024) discuss general issues regarding students' emotional needs regarding generative AI, emphasizing the role of lack of technology readiness and insufficient validation and integration in authentic educational contexts. For the preservation of students' emotions, which significantly influence student motivation, learning, and overall well-being (Fredrickson & Joiner, 2002), the emotional conditions surrounding the use of chatbots deserves investigation. This is further underscored by Brave and Nass (2007), as they emphasize the need for deep exploration into the effects on affect and emotion to aid in design improvements in human-computer interaction (HCI).

To examine the emotional impact of chatbot usage, we used a pre-post control study design. Students completed the same assignment, in which the intervention group was asked to complete the task with the assistance of a chatbot (i.e., ChatGPT), and students in the control group completed the task without that assistance. Assessments before and after completing the assignment included basic demographic questions, positive (PA) and negative affect (NA) scores, and an assessment of familiarity with chatbots.

This study contributes to the literature on AI chatbots in education by shifting attention from functional outcomes to emotional consequences. Recent scholarship has examined the widespread use of ChatGPT and similar tools in academic contexts, highlighting both potential benefits and risks (Memarian & Doleck, 2023; Stadler et al., 2024). While empirical work has identified behavioral predictors of chatbot adoption, including ease of use, usefulness, and trust, readiness and psychological outcomes remain underexamined (Hasan et al., 2024; Kleine et al., 2025). Related studies have shown that students with social anxiety may over-rely on AI companions, risking greater emotional distress in the long term (Hu et al., 2023). Our study addresses this gap by focusing specifically on positive and negative affect before and after chatbot usage and examining how familiarity with AI tools moderates this relationship. In doing so, we contribute to the growing field of affective computing and AI–human interaction in higher education. This contribution points directly to ensuring students and faculty alike can approach chatbot usage with the balance it deserves. More specifically, it showcases how the convenience afforded by chatbots may be negatively influencing students' positive affect.

2. Theory development

2.1. Theoretical foundation

The emotional impact of chatbot usage on students can be explored through the lens of Cognitive Offloading Theory (COT). COT refers to the tendency for individuals to delegate cognitive processes to external tools rather than engage in direct mental effort (Risko & Gilbert, 2016). The theory is closely linked to Extended Mind Theory (EMT) which argues that tools can become extensions of the mind (Clark & Chalmers, 1998). For example, Sparrow et al. (2011) found that individuals can offload memory to the internet by remembering how to find information instead of remembering the information itself. Offloading certain tasks can significantly reduce cognitive load, allowing students to allocate cognitive resources toward other activities. While this may make things more efficient, it can also lead to reduced engagement, diminished personal agency, and a lack of intrinsic reward from completing the given task (Risko & Dunn, 2015). Students who use chatbots to replace rather than supplement effort may experience a reduction in motivation and satisfaction as a result of being less personally invested (Fan et al., 2024; Zhai et al., 2021).

Self-Determination Theory (SDT) complements COT by explaining how basic psychological needs shape affective responses. SDT posits that individuals have a fundamental need for autonomy, competence, and relatedness (Deci & Ryan, 1985). When students feel in control, capable, and connected to the process, they exhibit higher motivation and engagement (Deci & Ryan, 2000). Conversely, if students feel a chatbot is simply completing their work for them, they may perceive a loss of agency and disconnection from the learning process (Anson, 2024). The extent to which students can effectively partner with a chatbot depends on their level of competence and familiarity with the technology. As such, familiarity with chatbots may be a critical factor in determining whether students experience engagement or detachment in response to chatbot usage.

Indeed, current research lacks explanation of the way students' chatbot usage influences their emotional wellbeing, which is problematic for contemporary learners since modern education programs not only allow AI use, but many times encourage its use. Therefore, this research aims to investigate the emotional effect chatbot usage has on students' academic work. We also consider the effect that students' familiarity with chatbots may have on their emotions. The sections below will outline the state of current research and hypothesize our expectations in this research.

2.2. AI chatbots

Chatbots are, arguably, the most common application of AI in education. The transformer-based architecture of these programs allows them to generate text that is fluent, coherent, and responsive to the requested context (Vaswani et al., 2017). Chatbots can perform various tasks including providing summaries of content, answering questions, and generating creative suggestions for copy. Such widespread applicability has made chatbots very popular with many students. Importantly, one primary characteristic which enhances chatbots is the multi-task learning capability. This capability enables a model to learn a given task with very little training via few-shot or zero-shot learning (Wei et al., 2022). Therefore, their flexibility and ease of deployment makes chatbots valuable tools that address the needs of many learners (Pinski & Benlian, 2024). Indeed, the appeal is obvious, as learners and educators can use chatbots as a means for intelligent tutoring, virtual assistants, or automated assessments. Chatbots provide students quick feedback and materials that are relevant in the manner that best suits them, while educators are able to spend more time engaging with their students in more profound and substantive ways (Memarian & Doleck, 2024).

Despite the widespread appeal, many issues in implementing chatbots into higher education remain unsolved. Data privacy, bias, cheating, and monopoly of AI tools underpin just a few issues stemming from under-governed areas of artificial intelligence (Popenici & Kerr, 2017; Memarian & Doleck, 2023; Carter et al., 2025). Additionally, over-reliance on chatbots can reduce learners' opportunity for the development of critical thinking (Bayne, 2015; Mogavi et al., 2023). Since chatbots currently have limited fact-checking abilities and provide factually incorrect outputs (Radford et al., 2019), they can generate ambiguous or erroneous responses, leading to users' frustration, anxiety, and disengagement (Bender et al., 2021). Importantly, the possibility of those frustrations increases without appropriate training, a problem poised to exacerbate inequality issues, which continue to be a major concern for AI usage in education (Holmes & Tuomi, 2022). In short, the opportunities presented may not always yield positive outcomes.

2.3. Student affect

Affect refers to the pleasant and unpleasant feelings someone experiences at a certain point in time (Wyer et al., 1999). Thus, affect is a general concept containing both positive and negative feelings used to explain a broad range of mood and emotion (Russell & Carroll, 1999). Positive Affect (PA) includes emotions such as joy, love, and contentment, while Negative Affect (NA) includes emotions like fear, anger, and sadness (Wyer et al., 1999). Though initially regarded as independent constructs (Diener & Emmons, 1984), research has shown a large degree of interplay between PA and NA (Folkman, 2008; Tugade & Fredrickson, 2004). Affect plays a critical role in shaping student engagement, motivation, and learning outcomes (Pekrun, 2006). Positive emotions such as curiosity, joy, and pride enhance cognitive processing and persistence. Those emotions facilitate deeper engagement with learning materials (Pekrun & Stephens, 2010). Conversely, negative emotions such as anxiety and frustration can impede learning by diverting cognitive resources away from task performance (Sweller, 1988).

The expansive adoption of technology likely has some effect on learners' affective states. However, strategies aimed to optimize learners' experiences remain underdeveloped, likely because the relationship between technologies and learners' affect remains largely unexplored (Holmes et al., 2023; Zhai et al., 2021), further supporting the need for the current research. Indeed, increased offerings of AI and digital learning tools have fundamentally shifted the student experience. For example, technology-enhanced learning (TEL) engages students both cognitively and emotionally, shaping their engagement, persistence, and satisfaction (D'Mello & Graesser, 2012). Within TEL though, chatbots open up a new paradigm of opportunities and challenges, both of which are discussed more deeply in the following.

Chatbots can provide personalized feedback, reduced cognitive load, and provide interactive learning experiences (Memarian & Doleck, 2023; Shute, 2008; Zhai et al., 2021). Indeed, chatbots can potentially enhance PA in students when the model and its capabilities align with students' individual needs. By offering feedback tailored to specific learning goals, chatbots improve motivation and bolster self-esteem, which deepen engagement and enjoyment (Hattie & Timperley, 2007; Shute, 2008; Vanichvasin, 2022). Chain-of-thought prompting can streamline critical thinking, helping students build confidence incrementally and experience positive emotions such as pride and curiosity (Wei et al., 2022). Interactive features spark excitement and interest (Luckin, 2017; Mayer, 2009), with creative affordances enabling students to brainstorm innovative ideas and engage in imaginative problem-solving (Beghetto & Kaufman, 2014). Additionally, chatbots reduce the cognitive load linked to repetitive or mundane tasks (Zhai et al., 2021), allowing students to focus on higher-order thinking and more meaningful coursework. These systems further contribute to social-emotional learning by simulating scenarios that nurture empathy, communication, and collaboration in safe, AI-driven environments (Holmes et al., 2023; Mogavi et al., 2023). Thus, the positive opportunities seem quite expansive.

Despite the benefits, chatbots also pose real concerns for affective states. Poorly designed systems or unrealistic expectations can result in frustration, confusion, or diminished self-efficacy (Kay et al., 2013; Wester et al., 2024). Gerlich (2025) argues that offloading to chatbots can actually reduce critical thinking skills. Technical limitations and algorithmic errors may misalign with students' emotional states or produce misleading outputs (Bender et al., 2021; Radford et al., 2019). Additionally, users' ambiguous prompts can entirely derail the learning process (Wei et al., 2022). Unfortunately, some feelings of inadequacy may stem from comparing one's work to AI-generated outputs (Mogavi et al., 2023). Importantly, over-reliance on chatbots can erode students' sense of agency and self-efficacy if they begin to doubt their own abilities without the assistance (Compeau & Higgins, 1995; Zhai et al., 2021; Lee, 2022). Additionally, using a chatbot may increase immediate success on a task but lead to less long-term understanding and learning (Grinschgl et al., 2021). Moreover, the rapid advancement of chatbots can spark uncertainty and anxiety about future job prospects (Pew Research Center, 2023; Tarafdar et al., 2011), as well as concerns about data privacy and bias (Holmes et al., 2023; Pasquale, 2015). Selim et al. (2024) thus describe chatbot usage as a double-edged sword that may simplify tasks and provide valuable resources and at the same time contribute to anxiety, dependence, and pessimism about the future.

Although chatbots excel in simulating human-like conversations, their lack of genuine empathy and emotional intelligence can lead to feelings of depersonalization (Wester et al., 2024). Nonetheless, they offer tailored feedback and immediate support (Pinski & Benlian, 2024). While Zhai et al. (2021) argue for chatbots' potential to enhance academic performance, students who become overly dependent risk missing both academic growth (Anson, 2024) and social growth, as interaction with technology can diminish face-to-face communication, causing isolation (Przybylski & Weinstein, 2013). Taken together, these findings underscore the complex interplay between the clear positives and negatives as they would relate to chatbot use in education. Although chatbots may help produce a better product, the experience is not always positive. Given the findings to date, and in conjunction with the previously outlined purpose of this research, we propose the following hypotheses.

H1. Students who use an AI chatbot to complete an assignment will report a greater decrease of PA compared to those who do not use an AI chatbot.

H2. Students who use an AI chatbot to complete an assignment will report a greater increase of NA compared to those who do not use an AI chatbot.

2.4. Familiarity

Beyond simply knowing the effect something may have on students, we need to understand how potentially negative effects might be limited. Therefore, and in line with SDT, the familiarity of chatbot usage possible explanatory value. Specifically, SDT suggests that competence reduces perceived difficulty and enhances perceived usefulness, thereby increasing positive emotional responses (Deci & Ryan, 1985). Said differently, students familiar with crafting prompts and interpreting chatbot outputs may experience greater satisfaction and motivation as their interactions yield productive results. Therefore, familiarity fosters a sense of mastery and control, contributing to PA through increased confidence and engagement. Bandura's (1997) self-efficacy theory posits that successful experiences build a sense of competence and optimism. As such, increasing familiarity with chatbots should serve to as a way to reduce barriers to effective use, allowing students to explore creative problem-solving and achieve academic goals (Wester et al., 2024). Although the effectiveness may help students' academic achievements (Zhai et al., 2021), the effect the tool has on emotional states is still unclear. Additionally, students who are well conversant with the AI tools may feel confident and satisfied, while those that are not well conversant with the technologies may end up being frustrated and overloaded (Pinski & Benlian, 2024), further reinforcing the need to consider preexisting familiarity with chatbots.

Regarding familiarity and its effect on students' perceptions, evidence suggests that students' perceptions of chatbots can influence their willingness to engage with these tools. While chatbots may enhance accessibility and motivation, challenges such as perceived complexity and lack of familiarity can lead to heightened anxiety and reduced engagement (Kleine et al., 2025). To a certain extent, chatbots encourage curiosity-driven exploration. Thus, students who understand the adaptive capabilities of chatbots are more likely to experiment with diverse prompts and use cases aligned with their academic interests. This exploratory behavior fosters intrinsic motivation and sustained engagement (Memarian & Doleck, 2024). By contrast, unfamiliarity with chatbots can exacerbate NA, such as frustration, anxiety, and confusion, stemming from technical challenges or unrealistic expectations. Students who lack familiarity may struggle with formulating prompts, interpreting outputs, or managing technical limitations, all of which have been linked to cognitive overload (Flavell, 1979; Venkatesh et al., 2003).

Familiarity also reduces emotional distress by fostering resilience through metacognitive awareness. For example, students who understand that chatbots are probabilistic tools with inherent limitations are less likely to feel disillusioned when encountering unexpected outputs (Zhai et al., 2021). Ethical awareness, as a component of familiarity, also helps students contextualize concerns about data privacy and algorithmic bias, promoting balanced perspectives (Pasquale, 2015). Finally, familiarity may temper social comparison effects, enabling students to view chatbot-generated content as learning aids rather than benchmarks for personal competence (Wester et al., 2024). Such findings suggest different responses to familiarity with chatbot usage may be expected. Therefore, in our effort to understand how we might limit the potential negative influence chatbot usage may facilitate, and in line with previous findings and our theoretical framework, we propose the following hypotheses.

H3. Familiarity with AI chatbots will moderate the relationship between AI chatbot usage and PA.

H4. Familiarity with AI chatbots will moderate the relationship between AI chatbot usage and NA.

3. Methodology

3.1. Research design

To examine the emotional impact of chatbot usage on students, this study employed a pre-post control group design. We compared an intervention group who completed an assignment using a chatbot (i.e., ChatGPT), and a control group who completed the same assignment without any assistance from a chatbot. This design evaluated emotional changes before and after the assignment, providing insights into the direct and indirect effects of chatbot usage. Participants were randomly assigned to either the control or intervention group at the start of the study. Both groups were instructed to analyze a business case study focused on negotiating a supplier contract, and craft a written proposal to submit to the company. The case study and assignment therein matched the difficulty level commensurate to the level of the course (i.e., 300-level undergraduate) in which the assignment was completed. The course was a requirement for general management students, and did not relate to using chatbots. Any experience with chatbots before the task would have been a result of students' use in ways unrelated to the course in which they completed the task used for this study. Instructions and guidelines remained the same for both groups, with only one exception. The control group was instructed not to use ChatGPT during the task, whereas the intervention group was explicitly instructed to use ChatGPT to assist with their work. To ensure all students had equal opportunity for assistance, participants in both groups had the same access to the instructor's advice.

Both groups completed a pre-assignment survey measuring demographic information, familiarity with chatbots, emotional intelligence, and baseline affect. After completing the assignment, participants completed a post-assignment survey, which reassessed their affect levels. In accordance with the institutional review board (IRB) at the university in which the sample occurred, participation was voluntary, and students could withdraw at any time without penalty. Anonymity and confidentiality were maintained throughout the study, and participants were offered additional academic support if they experienced distress due to the study tasks.

Participants were undergraduate students enrolled in a management course at a regional university. A total of 75 students participated in the study, with 38 assigned to the intervention group and 37 assigned to the control group. Participants provided informed consent before participating, and all data were collected in compliance with IRB guidelines. Students in both the intervention and control groups took on the role of small business suppliers negotiating a contract with Walmart. Students were tasked with drafting a professional contract proposal outlining pricing tiers, delivery timelines, and potential concessions to secure a favorable agreement while maintaining sustainable operations. The key difference between the groups was the instructed use of ChatGPT for the intervention group, who also received instructions to submit their prompts and chat-feedback in addition to the assignment deliverable. The control group received instruction not to use any AI tools and were notified that their submissions would be analyzed with an AI detector.

3.2. Measures

3.2.1. Affect

The Positive and Negative Affect Scale (PANAS: Watson et al., 1988) was used to assess both PA and NA. The scale consists of 20 items with 10 measuring positive emotions and 10 measuring negative emotions. Sample items include "interested," "distressed," "excited," and "upset." Responses were rated on a 5-point Likert scale, where 1 indicated "very slightly or not at all" and 5 indicated "extremely." Both positive and negative scales demonstrated excellent reliability, with α values of 0.847 (positive), and 0.849 (negative).

3.2.2. Familiarity

Following the approach of Kleine et al. (2025), participants reported their familiarity with chatbots on a 5-point Likert scale ranging from "very unfamiliar" to "very familiar." Specifically, we asked "Before completing the assignment, how familiar were you with chatbots, such as ChatGPT?" Familiarity was further validated by comparing it with students' self-reported frequency of chatbot usage.

3.3. Data analysis

Statistical analyses were conducted using IBM SPSS Statistics (Version 28). Normality was assessed using Kolmogorov-Smirnov and Shapiro-Wilk tests, and homogeneity of variance was evaluated with Levene's test, confirming that parametric tests were appropriate. Independent sample t-tests were used to assess baseline group equivalence and post-intervention differences in PA (PA) and NA (NA). Paired-sample t-tests examined within-group changes over time. Additionally, univariate analyses of variance (ANOVAs) were employed to test the moderating effect of familiarity with chatbots on affect changes, while controlling for age and gender (Derner et al., 2024; Memarian & Doleck, 2024).

4. Results

To ensure data reliability and validity, normality and homogeneity of variance were assessed prior to hypothesis testing. Kolmogorov-Smirnov and Shapiro-Wilk tests indicated that all variables approximated normality ($p > .05$). Levene's test confirmed no significant violations of variance equality for changes in PA ($F(1, 73) = 1.43$, $p = .236$) or NA ($F(1, 73) = 1.93$, $p = .169$), supporting the use of parametric tests. Independent-sample t-tests showed no significant differences between the intervention and control groups at baseline for PA ($t(73) = 0.07$, $p = .948$) or NA ($t(73) = -0.62$, $p = .538$), supporting the assumption of comparable groups prior to the intervention. Descriptive statistics for PA and NA before and after the intervention are reported in Table 1. Across the full sample ($N = 75$), the mean PA score decreased from 33.65 ($SD = 6.71$) to 31.80 ($SD = 7.69$), while the mean NA score increased from 20.68 ($SD = 6.77$) to 22.49 ($SD = 7.08$).

A paired-sample *t*-test revealed a significant decrease in PA from pre-intervention ($M = 33.65$, $SD = 6.71$) to post-intervention ($M = 31.80$, $SD = 7.69$), $t(74) = 2.07$, $p = .042$, $d = 0.24$, indicating a small effect size. An independent-sample *t*-test compared the change in PA between the intervention group (chatbot use) and the control group. Results showed a significantly greater reduction in PA for the intervention group ($M = -3.87$, $SD = 6.49$) compared to the control group ($M = 0.22$, $SD = 8.47$), $t(73) = 2.35$, $p = .011$, $d = 0.54$. Thus, H1 was supported.

A paired-sample *t*-test revealed a significant increase in NA from pre-intervention ($M = 20.68$, $SD = 6.77$) to post-intervention ($M = 22.49$, $SD = 7.08$), $t(74) = -2.21$, $p = .030$, $d = -0.26$. However, an independent-sample *t*-test comparing NA changes between the intervention and control groups found no significant difference, $t(73) = -0.46$, $p = .325$. Therefore, H2 was not supported.

A significant positive correlation was found between familiarity and usage frequency ($r = 0.566$, $p < .001$), suggesting consistency in how students perceived and reported their familiarity with chatbots. Additionally, cross-tabulation analyses demonstrated alignment between familiarity scores and frequency categories, with significant chi-square results ($\chi^2(16) = 57.43$, $p < .001$), further supporting the reliability. A univariate analysis of variance (ANOVA) examined whether familiarity with chatbots moderated the relationship between chatbot usage and changes in PA. As shown in Table 2, a significant interaction effect was found, $F(1, 71) = 4.50$, $p = .037$, $\eta^2 = 0.060$, indicating that familiarity had a moderate influence on the degree of PA change (Lakens, 2013). Fig. 1 illustrates this interaction effect, showing that higher familiarity is associated with reduced declines in PA for chatbot users. Therefore, H3 was supported. A similar ANOVA was conducted to assess whether familiarity moderated the relationship between chatbot usage and changes in NA. The interaction effect was not significant, $F(1, 71) = 0.046$, $p = .831$, $\eta^2 = 0.001$, suggesting that familiarity did not moderate this relationship. Therefore, H4 was not supported.

5. Discussion

This study explores the emotional impact of using chatbots on students' PA and NA and examined the moderating role of familiarity. The findings revealed a significant decrease in PA among students in the intervention group who used ChatGPT to complete the assignment, while those in the control group maintained stable PA levels. Similarly, although NA did increase overall, no significant differences emerged between the intervention and control groups. These findings highlight the nuanced emotional effects of chatbot usage, challenging the assumption that such tools inherently foster positive emotional experiences for students. Importantly, the use of chatbots for completion of an academic task had a significant impact on students' PA without a marked decrease of NA.

The observed decline in PA among the intervention group aligns with concerns that offloading cognitive tasks to chatbots may introduce challenges (Bender et al., 2021; Mogavi et al., 2023). Although the effect size for PA ($d = 0.24$) is considered small, this is consistent with what is commonly reported in educational and behavioral research. As Lipsey and Wilson (1993) note, small-to-moderate effect sizes are typical for instructional interventions. Moreover, even modest emotional shifts can be practically significant when scaled across student populations or sustained over time (Maher et al., 2013). While chatbots are celebrated for reducing repetitive tasks and providing personalized support (Zhai et al., 2021), this study suggests that their use may paradoxically reduce the intrinsic satisfaction of completing an assignment independently. This pattern aligns with recent findings that LLM-supported learners often experience reduced cognitive effort but also reduced engagement and depth in processing, suggesting that emotional cost may accompany cognitive ease (Stadler et al., 2024). Similarly, Memarian and Doleck (2023) argue that while chatbots offer personalized feedback and workload relief, they may also hinder the development of student autonomy and resilience when used without proper framing. On the one hand, the disconnect between expectations and actual chatbot output may contribute to this decline, as students unfamiliar with the nuances of prompting and interpreting chatbot responses may experience unmet expectations (Wei et al., 2022). On the other hand, it may be that students feel a reduction of PA due to lack of cognitive stimulation or agency. This aligns with research on cognitive offloading (Fan et al., 2024: Carter et al., 2024) which suggests that when users rely excessively on AI assistance, they may experience a loss of intrinsic motivation and engagement.

Interestingly, while the overall increase in NA was significant, changes in NA were not significantly different between group nor exhibit a moderating effect of familiarity. This may suggest that chatbots do not disproportionately increase negative emotions such as frustration or anxiety compared to traditional methods. It is also possible that students experienced frustration or fatigue during the task regardless of AI assistance, which would blunt any moderating effect of familiarity. This finding contrasts with prior literature, which argue chatbot-usage can lead to negative emotions, possibly stemming from errors or ambiguous outputs (Kay et al., 2013). While chatbot usage did not appear to cause

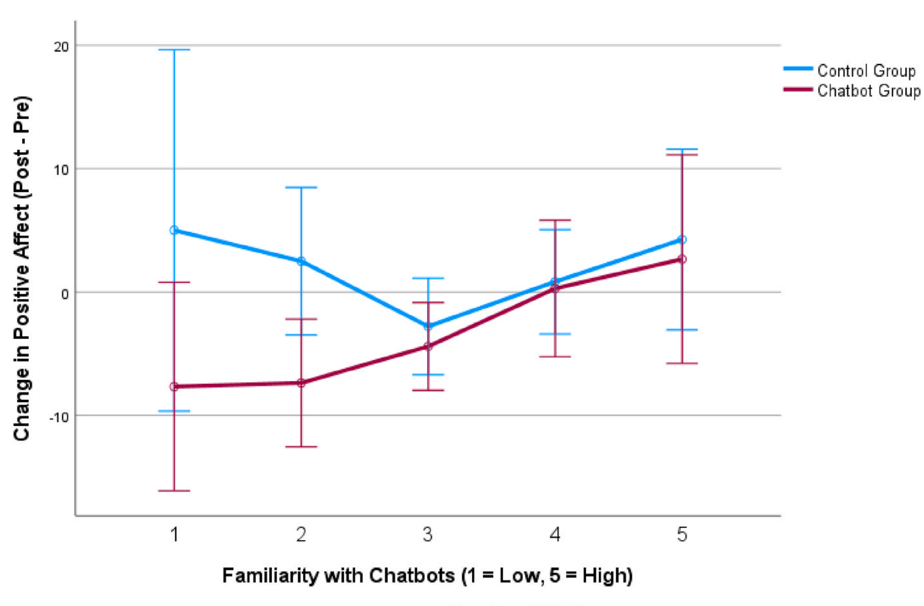

Fig. 1. Interaction between chatbot use and familiarity on positive affect change.

distress in students, it did appear to diminish the factors that contribute to positive learning experiences.

Familiarity with chatbots emerged as a significant moderator for PA changes, supporting the role of familiarity in shaping students' emotional responses. Students with greater familiarity experienced a less pronounced decline in PA, likely due to enhanced confidence and reduced cognitive demands associated with navigating the tool (Bandura, 1997; Venkatesh et al., 2003). This finding aligns with SDT, as familiarity fosters mastery (Kleine et al., 2025). It could therefore be argued that the results indicate students who were familiar with the technology could use it in a more positive way, while those unfamiliar with the technology did not necessarily have a negative experience, but were less likely to derive positive emotional benefits. Therefore, the effective and positive integration of chatbots in academic settings may depend significantly on users' proficiency and understanding of the technology.

These results have practical implications for integrating chatbots into educational settings. First, providing students with training and resources to develop familiarity with chatbots is essential for fostering positive emotional experiences. Given that the course in which the study occurred was a 300-level course, we suggest the need for a general education course earlier in students' course plans. That course could facilitate space in which educators should emphasize skill-building, such as crafting effective prompts and interpreting chatbot-generated outputs, to empower students and enhance their sense of agency. Additionally, and to extend educational research, future efforts could extend to graduate-level courses, where the students may already have more corporate training. Together, both earlier courses and investigations in more advanced courses could yield more precise effects as they relate to the training aimed at enhancing familiarity with chatbot use.

Practically, curriculum design should incorporate the strategic use of chatbots to complement traditional learning methods while students progressively build their proficiency and comfort with the technology. This will help ensure chatbots are used as tools for enrichment rather than as shortcuts that diminish agency and efficacy. Finally, educators themselves need adequate training and professional development to effectively integrate chatbots into their teaching practices. When teachers are well-versed in the capabilities, they can better support students by modeling effective usage (Bandura, 1997).

Though the context of our data remains limited to the educational setting, our results do bear potential significance for organizations as well. Indeed, chatbot use in work settings continues to rise, presumably due to the efficiencies it facilitates (Durach & Gutierrez, 2024). Like the education setting, users report mixed findings. While consumers' use of chatbots yields mostly negative reports (Chih et al., 2025; Huang & Dootson, 2022; Huang et al., 2024), reports on the employee perspective remain mostly limited to thoughts about chatbot usage. That is, some admit chatbot usage is “annoying” (Gkinko & Elbanna, 2022) and poses privacy and accuracy risks (Pillai et al., 2024). The effects of actual use, specifically on the employee's wellbeing, remain largely underexplored. We suggest a similar design to our pre-post study that may prove useful and further suggest the applicability of our theoretical framework. More specifically, the prevalence of SDT and COT in both educational organizational settings continues to be fragmented. Our extension into educational spaces helps extend theoretical boundaries, as the theoretical expectations in our hypotheses yielded positive results. This reinforces prior work on cognitive offloading, which suggests that delegating mental effort to AI tools may streamline task completion while reducing engagement and intrinsic reward (Risko & Gilbert, 2016; Stadler et al., 2024). Future research aimed at confirming the results of this study in organizational settings is necessary, as academic and industrial conditions differ. The findings contribute to cognitive offloading theory by highlighting that offloading not only affects memory or task complexity but also alters emotional engagement—suggesting a psychological trade-off that should be accounted for in future applications (Risko & Dunn, 2015; Stadler et al., 2024). Additionally, we extend self-determination theory by demonstrating that chatbot usage can either support or hinder autonomy and competence depending on the user's familiarity with the tool (Hasan et al., 2024; Kuhail et al., 2022). Practically, this suggests the importance of scaffolding not just technical skills, but emotional readiness. Institutions should treat prompt literacy, ethical use norms, and chatbot mindset as core onboarding modules (García-López et al., 2025).

6. Limitations

While this study provides valuable insights into the emotional impact of chatbot usage, several limitations must be acknowledged. First, the study involved a sample of 75 undergraduate students enrolled in a single 300-level management course at a single regional university. This restricts the findings' generalizability to other student populations, disciplines, and educational contexts. However, the use of validated measures and appropriate statistical analysis serves to enhance the reliability of findings (Lakens, 2002). The sample was sufficient enough to detect significant effects for key hypotheses, suggesting adequate statistical power for the primary analyses. Future research should replicate this study across a broader range of educational levels, course types, and institutions to assess the consistency and scalability of the observed emotional effects.

Additionally, the study used a short-term, task-specific intervention, which may not fully capture the long-term emotional and academic effects of chatbot usage. Extended studies examining repeated or prolonged engagement with chatbots could provide a more comprehensive understanding of their impact on student affect and psychological well-being. Additionally, the use of a single assignment may have limited the scope of emotional responses observed, as the nature of the task could influence the levels of cognitive demand and emotional engagement. While the study measured familiarity with chatbots as a potential moderator, other individual factors such as prior academic performance, technological proficiency, or baseline levels of self-efficacy were not assessed. These variables could influence both the emotional responses to chatbot usage and the overall outcomes, warranting further exploration. Another limitation lies in the self-reported nature of the affect measures, which are subject to social desirability bias and may not fully reflect students' true emotional states. While the PANAS scale is a well-validated tool, future research could incorporate physiological or behavioral measures of affect to triangulate self-reported data and enhance the robustness of findings.

7. Conclusion

This study provides important insights into the complex emotional impact chatbots may have on students. Findings revealed that, while chatbot usage may reduce PA, familiarity with these tools moderates the impact. The findings underscore the dual nature of chatbots. Although they offer immense potential to streamline academic tasks and reduce cognitive loads, they may also lead to diminished enjoyment for students who are less familiar. Importantly, training and support prove critical to maximizing the benefits of chatbots while simultaneously mitigating their emotional challenges. By addressing these challenges and leveraging familiarity as a key enabler, the academic landscape can include chatbots such that they not only enhance learning outcomes, but also promote emotional well-being, setting the stage for a balanced and effective use of AI in education.

CRediT authorship contribution statement

Justin W. Carter: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Justin T. Scott:** Writing – review & editing, Writing – original draft. **John D. Barrett:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis.

Declaration of generative AI in scientific writing

During the preparation of this work the authors used Chat GPT in order to improve clarity and writing of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table 1
Baseline Group Equivalence and Descriptive Statistics

Variable	Group	Mean (SD) or %	$t(df)/\chi^2(df)$	p
Age (years)	Intervention	21.45 (3.12)	0.34(73)	0.734
	Control	21.28 (2.98)		
Gender (% Female)	Intervention	56.80 %	0.48(1)	0.49
	Control	60.50 %		
PA (PA)	Intervention	33.71 (7.57)	0.07(73)	0.948
	Control	33.59 (5.80)		
NA (NA)	Intervention	21.11 (6.94)	−0.62(73)	0.538
	Control	20.24 (6.66)		
Familiarity	Intervention	3.87 (0.81)	−0.32(73)	0.751
	Control	3.91 (0.78)		

Table 2
ANOVA Results for Moderation Effects of Familiarity

Effect	SS	df	MS	F	p	η^2
PA Moderation						
Condition (Chatbot vs. Control)	292.692	1	292.692	6.479	0.013	0.084
Familiarity	570.836	1	570.836	12.636	<0.001	0.151
Condition × Familiarity	203.312	1	203.312	4.5	0.037	0.06
Error	3207.539	71	45.177			
Total	80217	74				
NA Moderation						
Condition (Chatbot vs. Control)	1.902	1	1.902	0.039	0.844	0.001
Familiarity	197.446	1	197.446	4.054	0.048	0.054
Condition × Familiarity	2.237	1	2.237	0.046	0.831	0.001
Error	3457.918	71	48.703			
Total	41661	74				

References

Anson, D. W. (2024). The impact of large language models on university students' literacy development: A dialogue with lea and Street's academic literacies framework. *Higher Education Research and Development*, 1–14. https://doi.org/10.1080/07294360.2024.2332259

Bandura, A. (1997). *Self-efficacy: The exercise of control.* Macmillan.

Bayne, S. (2015). Teacherbot: Interventions in automated teaching. *Teaching in Higher Education, 20*(4), 455–467. https://doi.org/10.1080/13562517.2015.1020783

Beghetto, R. A., & Kaufman, J. C. (2014). Classroom contexts for creativity. *High Ability Studies, 25*, 53–69. https://doi.org/10.1080/13598139.2014.905247

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big?. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610–623). https://doi.org/10.1145/3442188.3445922

Brave, S., & Nass, C. (2007). Emotion in human-computer interaction. In *The human-computer interaction handbook* (pp. 103–118). CRC Press.

Carter, J. W., Parnell, J. A., & Puckett, S. (2024). The role of psychological capital in shaping leaders' attitudes toward artificial intelligence: Mindset meets machine. *Journal of Business, Industry and Economics, 29*, 130–158.

Chan, C. K. Y., & Hu, W. (2023). Students' voices on generative AI: Perceptions, benefits, and challenges in higher education. *International Journal of Educational Technology in Higher Education, 20*(1), 43. https://doi.org/10.1186/s41239-023-00411-8

Chih, W.-H., Wang, K.-Y., & Banda, H. W. (2025). Chatbots at the frontline: Unveiling antecedents of customers' willingness to accept chatbot intervention in service recovery. *Journal of Retailing and Consumer Services, 84*, Article 104254. https://doi.org/10.1016/j.jretconser.2025.104254

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis, 58*(1), 7–19. https://doi.org/10.1093/analys/58.1.7

Compeau, D. R., & Higgins, C. A. (1995). Computer self-efficacy: Development of a measure and initial test. *MIS Quarterly, 19*(2), 189–211. https://doi.org/10.2307/249688

Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. New York, NY: Plenum.

Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry, 11*(4), 227–268. https://doi.org/10.1207/S15327965PLI1104_01

Derner, E., de la Fuente, S. S., Gutiérrez, Y., Moreda, P., & Oliver, N. (2024). Leveraging large language models to measure gender bias in gendered languages. *arXiv preprint arXiv:2406.13677*. https://doi.org/10.48550/arXiv.2406.13677

Diener, E., & Emmons, R. A. (1984). The Independence of positive and NA. *Journal of Personality and Social Psychology, 47*(5), 1105–1117. https://doi.org/10.1037/0022-3514.47.5.1105

D'Mello, S. K., & Graesser, A. C. (2012). Dynamics of affective states during complex learning. *Learning and Instruction, 22*(2), 145–157. https://doi.org/10.1016/j.learninstruc.2011.10.001

Durach, C. F., & Gutierrez, L. (2024). "hello, this is your AI co-pilot"–operational implications of artificial intelligence chatbots. *International Journal of Physical Distribution & Logistics Management, 54*(3), 229–246.

Fan, Y., Tang, L., Le, H., Shen, K., Tan, S., Zhao, Y., Shen, Y., Li, X., & Gašević, D. (2024). Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance. *British Journal of Educational Technology*, 1–42. https://doi.org/10.1111/bjet.13544, 00.

Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist, 34*(10), 906–911. https://doi.org/10.1037/0003-066X.34.10.906

Folkman, S. (2008). The case for positive emotions in the stress process. *Anxiety, Stress & Coping, 21*(1), 3–14. https://doi.org/10.1080/10615800701740457

Fredrickson, B. L., & Joiner, T. (2002). Positive emotions trigger upward spirals toward emotional well-being. *Psychological Science, 13*(2), 172–175. https://doi.org/10.1111/1467-9280.00431

García-López, I. M., González, C. S. G., Ramírez-Montoya, M. S., & Molina-Espinosa, J. M. (2025). Challenges of implementing ChatGPT on education: Systematic literature review. *International Journal of Educational Research Open, 8*, Article 100401.

Gerlich, R. N. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. *Societies, 15*(1), 6. https://doi.org/10.3390/soc15010006

Gkinko, L., & Elbanna, A. (2022). Hope, tolerance and empathy: Employees' emotions when using an AI-enabled chatbot in a digitalised workplace. *Information Technology & People, 35*(6), 1714–1743.

Grinschgl, S., Papenmeier, F., & Meyerhoff, H. S. (2021). Consequences of cognitive offloading: Boosting performance but diminishing memory. *Quarterly Journal of Experimental Psychology, 74*(9), 1477–1496. https://doi.org/10.1177/17470218211008060

Hasan, M. R., Chowdhury, N. I., Rahman, M. H., Syed, M. A. B., & Ryu, J. (2024). Understanding AI chatbot adoption in education: PLS-SEM analysis of user behavior factors. *Computers in Human Behavior: Artificial Humans, 2*(2), Article 100098. https://doi.org/10.1016/j.chbah.2024.100098

Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research, 77*(1), 81–112. https://doi.org/10.3102/003465430298487

Holmes, W., Bialik, M., & Fadel, C. (2023). Artificial intelligence in education: Promises and implications for teaching and learning. *Center for Curriculum Redesign*.

Holmes, W., & Tuomi, I. (2022). State of the art and practice in AI in education. *European Journal of Education, 57*(4), 542–570. https://doi.org/10.1111/ejed.12533

Hu, B., Mao, Y., & Kim, K. J. (2023). How social anxiety leads to problematic use of conversational AI: The roles of loneliness, rumination, and mind perception. *Computers in Human Behavior, 145*, Article 107760. https://doi.org/10.1016/j.chb.2023.107760

Huang, Y. S. S., & Dootson, P. (2022). Chatbots and service failure: When does it lead to customer aggression. *Journal of Retailing and Consumer Services, 68*, Article 103044. https://doi-org.ezproxy.una.edu/10.1016/j.jretconser.2022.103044.

Huang, D., Markovitch, D. G., & Stough, R. A. (2024). Can chatbot customer service match human service agents on customer satisfaction? An investigation in the role of trust. *Journal of Retailing and Consumer Services, 76*. N.PAG https://doi-org.ezproxy.una.edu/10.1016/j.jretconser.2023.103600.

Kay, J., Reimann, P., Diebold, E., & Kummerfeld, B. (2013). MOOCs: So many learners, so much potential. *IEEE Intelligent Systems, 28*(3), 70–77. https://doi.org/10.1109/MIS.2013.66

Kleine, A. K., Schaffernak, I., & Lermer, E. (2025). Exploring predictors of AI chatbot usage intensity among students: Within-and between-person relationships based on the technology acceptance model. *Computers in Human Behavior: Artificial Humans, 3*, Article 100113. https://doi.org/10.1016/j.chbah.2024.100113

Kuhail, M. A., Alturki, N., Alramlawi, S., & Alhejori, K. (2022). Interacting with educational chatbots: A systematic review. *Education and Information Technologies, 28*(1), 973–1018.

Labadze, L., Grigolia, M., & Machaidze, L. (2023). Role of AI chatbots in education: Systematic literature review. *International Journal of Educational Technology in Higher Education, 20*(56), 1–17.

Lipsey, M. W., & Wilson, D. B. (1993). The efficacy of psychological, educational, and behavioral treatment: Confirmation from meta-analysis. *American Psychologist, 48*(12), 1181–1209. https://doi.org/10.1037/0003-066X.48.12.1181

Luckin, R. (2017). Towards artificial intelligence-based assessment systems. *Nature Human Behaviour, 1*(3), 28. https://doi.org/10.1038/s41562-016-0028

Maher, J. M., Markey, J. C., & Ebert-May, D. (2013). The other half of the story: Effect size analysis in quantitative research. *CBE-Life Sciences Education, 12*(3), 345–351. https://doi.org/10.1187/cbe.13-04-0082

Mayer, R. E. (2009). *Multimedia learning: Principles and applications* (2nd ed.). Cambridge University Press.

Memarian, B., & Doleck, T. (2023). ChatGPT in education: Methods, potentials and limitations. *Computers in Human Behavior: Artificial Humans*. , Article 100022. https://doi.org/10.1016/j.chbah.2023.100022

Memarian, B., & Doleck, T. (2024). Human-in-the-loop in artificial intelligence in education: A review and entity-relationship analysis. *Computers in Human Behavior: Artificial Humans, 2*, Article 100053. https://doi.org/10.1016/j.chbah.2024.100053

Mogavi, R. H., Deng, C., Kim, J. J., Zhou, P., Kwon, Y. D., Metwally, A. H. S., Tlili, A., Bassanelli, S., Bucchiarone, A., Gujar, S., Nacke, L. E., & Hui, P. (2023). ChatGPT in education: A blessing or a curse? A qualitative study exploring early adopters' utilization and perceptions. *Computers in Human Behavior: Artificial Humans, 2*, Article 100027. https://doi.org/10.1016/j.chbah.2023.100027

Ortega-Ochoa, E., Sabaté, J. M., Arguedas, M., Conesa, J., Daradoumis, T., & Caballé, S. (2024). Exploring the utilization and deficiencies of generative artificial intelligence in students' cognitive and emotional needs: A systematic mini-review. *Frontiers in Artificial Intelligence, 7*, Article 1493566.

Pasquale, F. (2015). *The black box society: The secret algorithms that control money and information*. Harvard University Press.

Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review, 18*(4), 315–341. https://doi.org/10.1007/s10648-006-9029-9

Pekrun, R., & Stephens, E. J. (2010). Achievement emotions: A control-value approach. *Social and Personality Psychology Compass, 4*(4), 238–255. https://doi.org/10.1111/j.1751-9004.2010.00259

Pew Research Center. (2023). *What the data says about americans' views of artificial intelligence*. Pew Research Center. https://www.pewresearch.org/short-reads/2023/11/21/what-the-data-says-about-americans-views-of-artificial-intelligence/.

Pillai, R., Ghanghorkar, Y., Sivathanu, B., Algharabat, R., & Rana, N. P. (2024). Adoption of artificial intelligence (AI) based employee experience (EEX) chatbots. *Information Technology & People, 37*(1), 449–478.

Pinski, M., & Benlian, A. (2024). AI literacy for users: A comprehensive review and future research directions of learning methods, components, and effects. *Computers in Human Behavior: Artificial Humans, 2*, Article 100062. https://doi.org/10.1016/j.chbah.2024.100062

Popenici, S., & Kerr, S. (2017). Exploring the impact of artificial intelligence on teaching and learning in higher education. *Research and Practice in Technology Enhanced Learning, 12*(1), 1–13. https://doi.org/10.1186/s41039-017-0062-8

Przybylski, A. K., & Weinstein, N. (2013). Can you connect with me now? How the presence of Mobile communication technology influences face-to-face conversation quality. *Journal of Social and Personal Relationships, 30*(3), 237–246. https://doi.org/10.1177/0265407512453827

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI Blog, 1*(8), 9.

Risko, E. F., & Dunn, T. L. (2015). Storing information in-the-world: Metacognition and cognitive offloading in a short-term memory task. *Consciousness and Cognition, 36*, 61–74. https://doi.org/10.1016/j.concog.2015.05.014

Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences, 20*(9), 676–688. https://doi.org/10.1016/j.tics.2016.07.002

Russell, J. A., & Carroll, J. M. (1999). On the bipolarity of positive and NA. *Psychological Bulletin, 125*(1), 3–30. https://doi.org/10.1037/0033-2909.125.1.3

Selim, R., Basu, A., Anto, A., Foscht, T., & Eisingerich, A. B. (2024). Effects of large language model–based offerings on the well-being of students: Qualitative study. *JMIR Formative Research, 8*, Article e64081. https://doi.org/10.2196/64081

Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research, 78*(1), 153–189. https://doi.org/10.3102/0034654307313795

Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory: Cognitive consequences of having information at our fingertips. *Science, 333*(6043), 776–778. https://doi.org/10.1126/science.1207745

Stadler, M., Bannert, M., & Sailer, M. (2024). Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior, 160*, Article 108386. https://doi.org/10.1016/j.chb.2024.108386

Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science, 12*(2), 257–285. https://doi.org/10.1207/s15516709cog1202_4

Tarafdar, M., Tu, Q., Ragu-Nathan, B. S., & Ragu-Nathan, T. S. (2011). Crossing to the dark side: Examining creators, outcomes, and inhibitors of technostress. *Communications of the ACM, 54*(9), 113–120. https://doi.org/10.1145/1995376.199540

Tugade, M. M., & Fredrickson, B. L. (2004). Resilient individuals use positive emotions to bounce back from negative emotional experiences. *Journal of Personality and Social Psychology, 86*(2), 320–333. https://doi.org/10.1037/0022-3514.86.2.320

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems, 30*, 5998–6008.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly, 27*(3), 425–478. https://doi.org/10.2307/30036540

Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of personality and social psychology, 54*(6), 1063. https://doi.org/10.1037/0022-3514.54.6.1063

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain of thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems, 35*, 24824–24837.

Wester, J., de Jong, S., Pohl, H., & van Berkel, N. (2024). Exploring people's perceptions of LLM-Generated advice. *Computers in Human Behavior: Artificial Humans, 2*, Article 100072. https://doi.org/10.1016/j.chbah.2024.100072

Wyer, R. S., Clore, G. L., & Isbell, L. M. (1999). Affect and information processing. In M. P. Zanna (Ed.), *Advances in experimental social psychology* (Vol. 31, pp. 1–77). Academic Press. https://doi.org/10.1016/S0065-2601(08)60271-3.

Yan, L., Sha, L., Zhao, L., Li, Y., Martinez-Maldonado, R., Chen, G., … Gašević, D. (2024). Practical and ethical challenges of large language models in education: A systematic scoping review. *British Journal of Educational Technology, 55*(1), 90–112.

Zhai, X., Chu, X., Chai, C. S., Jong, M. S. Y., Istenic, A., Spector, M., Liu, J.-B., Yuan, J., & Li, Y. (2021). A review of artificial intelligence (AI) in education from 2010 to 2020. *Complexity*. , Article 8812542. https://doi.org/10.1155/2021/8812542